

Stereotype Lift and Stereotype Threat Effects on Subgroup Mean Differences for Cognitive Tests: A Meta-Analysis of Adult Samples

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A large body of literature has studied the effect of stereotype threat and stereotype lift on cognitive test performance. Research on stereotype threat (ST) examines whether the awareness of a negative stereotype can decrease stereotyped group members' test performance. A less commonly studied influence of stereotypes is stereotype lift (SL), defined as an increase in a group's test performance due to not being part of a negative stereotype. For example, men might perform better on math tests if they are primed on the stereotype that men are better than women at math. Walton and Cohen (2003) previously meta-analyzed the impact of SL on cognitive tests, finding an overall $d = 0.24$. We report an updated meta-analysis on SL with more samples and moderator analyses. We then meta-analyzed between-group effects (majority–minority group differences both in the presence and absence of SL and ST) to compare their relative contributions to subgroup mean differences on cognitive tests. Our results indicate that SL has a small influence on cognitive test performance ($d = 0.09$, $SD_{res} = 0.19$), and that subgroup mean differences result largely from between-group effects rather than from the effects of ST and SL.

Keywords: stereotype lift, stereotype threat, cognitive ability

A large body of research literature has examined the effect of stereotype threat (ST) on cognitive test performance. Steele and Aronson's (1995) landmark work introduced the construct ST, defined as a situation in which awareness of a negative stereotype about a person's group causes that person to be concerned about being judged or treated negatively based on this stereotype (Spencer et al., 2016). For example, Steele and Aronson, and much of the related literature, examined the stereotype that men are better than women at math. Stereotype threat theory suggests that if a woman is aware of this stereotype, she may then feel threatened during a math test. The experience of threat consumes cognitive resources that would otherwise be available to focus on the test, thus decreasing test performance.

Self-categorization theory (Turner & Reynolds, 1987) notes that everyone balances unique identities as individuals in addition to social identities as members of social groups. An assumption of this theory is that people are implicitly motivated to retain positive and avoid negative social identities. Negative stereotypes can be conceptualized as a threat to one's social identity because they often involve a comparison between two subgroups within a given social identity (Schmader et al., 2001). The identity engagement model (Cohen & Garcia, 2008) illustrates this process. For ST, it posits that a primed social identity threatens performance by inducing concern about possibly confirming the stereotype linked to that social identity.

A number of mechanisms underlying the effects of ST have been proposed and supported. Schmader (2010) summarized support for mechanisms including reduced working memory capacity, increased uncertainty about one's abilities, and increased anxiety and hypervigilance about the possibility of one's performance confirming the stereotype. These mechanisms result in a reduction of the cognitive resources that a test-taker can devote to completing the test, thus leading to reduced test performance. An alternative mechanism for ST effects in laboratory settings is that threat reduces test-taking motivation, leading to low effort and low performance (Shewach et al., 2019). In operational settings, there are clear incentives for investing effort to maximize test performance (e.g., to succeed in getting the job for which one is applying). But in the common setting of college students participating in experiments, there are generally no consequences linked to test performance. Thus, minimizing effort in response to a threat manipulation is a plausible alternate explanation for observed differences in performance in threat and nonthreat conditions. A meta-analysis by Shewach et al., to be treated in detail later, found that the threat effect is reduced by more than 50% when incentives for test performance (e.g., payment for meeting a performance standard) are included in threat experiments, lending support to motivation as a contributing factor to threat effects in lab settings.

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Instead of a performance reduction, stereotype lift (SL) is a boost in performance for members of a nonstereotyped group when comparing their performance to stigmatized group members' performance (Walton & Cohen, 2003). For example, men might experience a boost in performance on a math test when they are aware of the stereotype that men are better than women at math. The nonnegatively stereotyped group members can compare themselves to the stereotyped group members, enjoy a comparatively positive social identity, which increases their self-efficacy (Bandura, 1986), maintains self-confidence and motivation, and limits anxiety and self-doubt, thus aiding performance (Walton & Cohen, 2003).

A Framework for Examining Stereotype Effects

The above two mechanisms of ST and SL are posited as influences on performance in the stereotyped domain, and the present study focuses on the domain of performance on cognitive tests. Our interest is in the use of cognitive tests for high-stakes decisions in settings, such as admissions in higher-education contexts and employee selection in work contexts. Thus, we limit our investigation to ST and SL effects in adult samples, acknowledging that there is also extensive work on stereotype effects in child samples (Flore & Wicherts, 2015).

A source of potential confusion is that treatments of stereotype effects vacillate between discussion of within-group and between-group effects. Figure 1 presents a framework for a clear understanding of these effects. It presents four cells of a 2 × 2 framework, with membership in a stereotyped group versus nonmembership in a stereotyped group as one dimension and being tested in a manipulated

threat condition versus a nonthreat condition as the second dimension. Comparison of stereotyped group mean performance in threat versus nonthreat conditions (cell A vs. cell B) is a within-group comparison, which indexes the effects of ST. Comparison of nonstereotyped group mean performance in threat versus nonthreat conditions (cell C vs. cell D) is a within-group comparison, which indexes the effects of SL. Thus, in experimental settings, a single manipulation can be used to examine both ST and SL: a threat manipulation is posited to create both ST in the stereotyped group and SL in the nonstereotyped group.

In Figure 1, two between-group comparisons are of interest: stereotyped group versus nonstereotyped group mean differences in conditions with ST and SL manipulations present versus conditions without ST and SL manipulation. Figure 1 illustrates comparisons, and the text below elaborates.

Comparison 1: Previous Meta-Analytic Work on ST Effects

There is a wealth of research on ST effects. Some of these studies examined noncognitive test performance, such as athletes "choking under pressure" due to ST (Beilock & McConnell, 2004; Stone et al., 2012). Most studies, though, examined performance on cognitive tests. The most recent meta-analyses are those by Shewach et al. (2019), focusing on adult samples, and Flore and Wicherts (2015), focusing on child samples. They found overall mean ST effects of $d = -0.31$ ($SD_{res} = 0.38$) and $d = -0.22$ ($SD_{res} = 0.32$), respectively. Importantly, Shewach et al. noted that ST research comprises almost exclusively laboratory studies in low-stakes settings. As a result,

Figure 1

Comparisons of Interest in Stereotype Threat and Stereotype Lift Research and Current Status of Knowledge of Each Comparison

Group studied	Threat condition	Nonthreat condition
Stereotyped group members	A	B
Non-stereotyped group members	C	D

Note. Text is included below to further explain this figure.

Two Within-Group Comparisons:

Comparison 1: Stereotype Threat: Within-group comparison of conditions A and B: comparing stereotyped group performance in threat vs. nonthreat conditions

Shewach et al.'s (2019) meta-analysis finds an overall mean threat effect of -0.31 SD , reducing to -0.14 SD in lab studies with conditions that are plausible in operational testing settings, to -0.01 in operational testing settings, and to -0.14 in lab studies with a motivational incentive for test performance.

Comparison 2: Stereotype Lift: Within-group comparison of conditions C and D: comparing non-stereotyped group performance in threat vs. nonthreat conditions

Walton and Cohen's (2003) meta-analysis finds an overall lift effect of 0.24 SD . Paralleling the Shewach et al. (2019) ST meta-analysis, the present study updates this to a) include more recent studies, b) limit the analysis to adult samples, and c) exclude studies making the common error of adjusting for a prior test when computing a d value. See Shewach et al. for details on this error.

Two Between-Group Comparisons:

Comparison 3: Subgroup differences in the presence of manipulated threat: comparing conditions A and C.

Comparison 4: Subgroup difference in the absence of manipulated threat: comparing conditions B and D.

The difference between Comparison 3 and Comparison 4 subgroup differences indexes the joint effects of ST and SL on subgroup differences. There is a large amount of published work erroneously asserting that between-group differences reduce to zero in non-threat conditions, initially documented by Sackett et al (2004), with the pattern of mischaracterization continuing to the present (Tomeh and Sackett, 2022). The present study is the first to systematically examine Comparison 3 and 4 between-group subgroup differences.

they cautioned against generalizing these lab studies' overall findings to operational test use in high-stakes settings, such as personnel selection. They subsequently focused on a subsample of studies with features that are plausible in high-stakes settings. First, they excluded manipulations that could not be used in an employment context. These would include manipulations that are explicit about mean score differences, such as "the test you are about to take is one on which members of your group score lower on average" or mention mean differences without specifying which group scores higher. Prior work has argued that it would be unethical to call attention to group differences as part of test instructions in operational settings (Sackett & Ryan, 2011).

Second, they included only studies using a no-threat control group as a comparison group, as opposed to a threat elimination or threat reversal comparison group (e.g., falsely stating there are no group differences on the test in a threat elimination setting or falsely stating that the lower scoring group in fact scores higher in a threat reversal setting). Neither of these could be used in an operational setting for the same ethical reasons as above (i.e., one cannot ethically give false information to applicants).

Third, they only included studies in which tests were scored using a number correct metric. A "percent correct among items attempted" metric is used with some frequency in threat research but would not be used in operational settings, as such tests are readily coached by telling test takers to only give a response if they are confident that it is correct.

After narrowing their focus, Shewach et al. (2019) produced a much weaker estimate of threat effects using features that are feasible in operational settings ($d = -0.14$, $SD_{res} = 0.31$). They also found a comparable mean effect of $d = -0.14$ in settings with motivational incentives for high test performance and a mean effect of $d = -0.01$ in the small number of studies ($k = 4$) in actual operational settings. They thus concluded that ST effects in operational settings are likely to be smaller than the effects commonly found in laboratory settings.

Comparison 2: Previous Meta-Analytic Work on SL Effects

Walton and Cohen (2003) meta-analyzed studies examining SL effects on cognitive tests. They focused on the literature's most commonly studied social identities, namely, race and gender. Additionally, they chose to examine only studies in which a stereotype-present condition could be compared to a condition where that stereotype was removed. For example, they examined studies where participants were either told (a) that men are better than women at math or (b) that men and women perform equally well at math. Across these studies, they calculated a mean SL effect of $d = 0.24$.

We offer four reasons for revisiting SL effects. The first is simply the availability of a considerably larger number of studies, given the passage of time since Walton and Cohen's (2003) meta-analysis. Second, SL effects can be estimated by various comparisons: between a "lift" group and (a) a no-information control group (e.g., "please take this test"), (b) a threat-removal group (e.g., "there are no mean differences on the test you are about to take"), or (c) a reversal group (e.g., "on this test, the negatively stereotyped group actually performs better"). Walton and Cohen focused only on the comparison condition in which a negative stereotype is

removed (i.e., a threat-removal group). As discussed in the meta-analysis by Shewach et al. (2019), focusing solely on stereotype removal instructions does not provide an analog to operational testing settings, as it is not ethically feasible to tell test takers that the test they are taking does not show group differences when in fact it does. Shewach et al. also found a weaker ST effect for studies using no-threat control conditions as the comparison group ($d = -0.28$, $SD_{res} = 0.33$) in which the negative stereotype is not mentioned, as compared to using removal conditions as the comparison group ($d = -0.39$, $SD_{res} = 0.47$). Therefore, we argue that it is useful to investigate whether the SL effect size differs across control, removal, and reversal comparison groups.

Third, Walton and Cohen (2003) included many studies that used means and SD s that were adjusted for performance on a prior cognitive test, as did many ST studies in the Shewach et al. (2019) meta-analysis. A study may, for example, report means and SD s on Scholastic Aptitude Test math questions administered in the experimental setting that are adjusted by the participants' prior Scholastic Aptitude Test math performance. As explained by Shewach et al., adjusting scores based on prior performance creates an upwardly biased d value. Using a prior score as a covariate is a longstanding research practice, and we are not arguing against its usefulness for some purposes. Experimental research controls for covariates to increase the power of statistical tests. However, when estimating standardized differences between two groups, one's interest is the magnitude of difference in unadjusted SD units. If a test has a SD of 10, then a mean difference of 5 can be interpreted as indicating that the two groups being compared differ by 0.5 SD units. If one adjusted the SD by removing variance shared with a covariate (e.g., a prior test score), then the resulting adjusted SD would be smaller, reflecting only variance not shared with the prior test. In the above example, dividing the mean difference of 5 by this smaller adjusted SD results in an overestimate of the difference between the two groups. Thus, Shewach et al. excluded such studies with biased effect size estimates from their meta-analysis.

Reviewing Walton and Cohen's (2003) meta-analysis, 11 out of 43 (25.6%) samples could be identified as using adjusted means and SD s. Twenty samples did not use adjustments, and 11 samples could not be found or evaluated. Thus, it would be useful to estimate SL effects using only studies which report effects that are not adjusted for prior test performance.

Fourth, Walton and Cohen's (2003) study included both adult and nonadult samples, which is reasonable as they focused on people in general settings compared to our interest in adults in higher education and workplace settings. Adult populations more often experience high-stakes testing environments that use cognitive tests. Thus, an estimate of SL effects in adult samples would be useful.

Comparisons 3 and 4: Between-Group Comparisons in the Presence or Absence of Manipulated Threat

While many inquiries into ST or SL focus on the within-group effects discussed above (e.g., By what magnitude do mean scores of a stereotyped group differ in threat vs. nonthreat conditions?), inquiries also often focus on between-group effects, attempting to answer questions like "what is the magnitude of White-Black mean test score differences in the absence of ST?" There is confusion in the literature about ST effects on between-group mean differences. Sackett et al. (2004) noted that the findings of Steele and Aronson's (1995)

initial set of studies on ST were widely misinterpreted. They examined how Steele and Aronson's findings were reported in journal articles, psychology textbooks, and the popular press. Sackett et al. found that studies focused on between-group effects were frequently described inaccurately, often with assertions that subgroup mean differences were present in threat conditions yet completely absent in control conditions. Sackett et al. demonstrated that many authors misinterpreted the fact that Steele and Aronson controlled for performance on a prior cognitive test. These authors thus transformed correct statements such as "controlling for prior performance, mean differences in a threat condition disappear when threat is removed" into incorrect statements like "mean differences in a threat condition disappear when threat is removed." Sackett et al. reported that 91% of journal articles, 67% of psychology textbooks, and 88% of popular press articles mischaracterized Steele and Aronson in this way.

Tomeh and Sackett (2022) investigated whether more recent work avoided this error in interpretation. They focused on journal articles and textbooks, finding that 63% of journal articles and 41% of textbooks still mischaracterize the results of Steele and Aronson (1995). A glass-half-full perspective notes that this is an improvement; nonetheless, misinterpretation continues at a substantial rate.

These reviews (Sackett et al., 2004; Tomeh & Sackett, 2022) cautioned against the misinterpretation that group mean differences reduce to zero in nonthreat conditions. But they did not offer a systematic investigation of the joint effects of ST and SL on subgroup mean differences. We offer such an investigation in this article.

One potential response is that one can answer this question of the effects of ST and SL on subgroup mean differences by pooling results from separate inquiries that have already been conducted, and thus the answer to this question is already known. First, as noted above, meta-analyses of within-group ST studies report an overall mean effect of about $-0.31 SD$ across all studies and a smaller estimate of about $-0.14 SD$ in studies with features feasible in operational test settings (Shewach et al., 2019). Second, also as noted above, a meta-analysis of within-group SL studies reports a mean effect of $0.24 SD$ (Walton & Cohen, 2003). Third, there is extensive research on subgroup differences on cognitive ability tests, with the value of $1.0 SD$ commonly offered as the mean White-Black difference and $0.67 SD$ as the mean White-Hispanic difference (Sackett et al., 2008). If one assumes that these subgroup differences are obtained in settings in which ST and SL are present (as is posited to be the case for high-stakes testing), one can combine the above three pieces of information. If majority test takers on average have scores inflated by $0.24 SD$ due to SL and minority test takers on average have scores lowered by $0.14 SD$ (the mean SD effect in studies with features feasible in operational settings), then $0.38 SD$ of the $1.0 SD$ White-Black difference is attributable to threat effects. A residual difference of $0.62 SD$ remains after removing ST and SL effects. By this logic, one would conclude that ST and SL account for about one third of the observed White-Black difference on cognitive ability tests.

We argue, however, that the above approach is not persuasive. It pools three separate meta-analytic estimates and carries the implicit assumption that the studies going in the analyses of ST effects, SL effects, and subgroup differences either use comparable tests and comparable samples or that the choice of test or sample is irrelevant (Sackett et al., 2017). We see it as quite possible that both samples

and tests differ in ST and SL studies from those used to derive estimates of mean score differences.

With regard to samples, Roth et al.'s (2001) meta-analysis of White-Black mean differences on cognitive tests produced a wide variety of means, depending on the sample used. For example, job applicant samples produce a mean White-Black difference of $1.0 SD$. Incumbent samples are commonly considerably smaller, with estimates in the 0.4 – 0.6 range produced in a number of analyses. At the other extreme, military applicant samples produce a mean difference of $1.46 SD$. Clearly, the sample studied has a large effect. Additionally, the proportion of the total subgroup difference attributable to stereotype effects changes depending on the mean differences (0.5 vs. 1.0 vs. 1.46). With regard to tests, most ST and SL research is done in laboratory settings, and it is common to make use of a relatively short test. Smaller subgroup differences may be found simply as a result of lower test reliability.

Sackett et al. (2017) proposed a solution to this problem, which they termed the "head-to-head comparison." If addressing a research inquiry requires multiple pieces of information, they call for obtaining each of the pieces of information from each of the samples used and using the same measures (in this case, tests). This avoids the potential confounding effects of these differences in sample and tests used.

We propose using the head-to-head comparison strategy here. Analysis can be limited to samples where a subgroup mean difference estimate can be obtained both in conditions where ST and SL effects are expected (e.g., a condition in which threat is induced by manipulation) and where ST and SL effects are not expected (e.g., a condition in which threat is not induced). With participants randomly assigned to conditions, the joint effects of ST and SL on mean differences can be obtained by comparing the results' estimates of mean differences in the presence or absence of manipulated stereotype effects.

Study 1: A New Meta-Analysis of SL Effects

For the reasons noted above, it is better to regard the present work as a separate examination of SL rather than an update to Walton and Cohen's previous meta-analysis, as we differ from their study with regards to the choice of comparison groups analyzed, statistics considered, and samples examined. Our first research question is thus:

Research Question 1: What is the effect of SL on cognitive test performance in adult samples?

In addition to exploring an overall SL effect, we also examine the influence of six moderators. The following research questions address these moderators.

Moderator 1: Comparison Group Type

A stereotype can be communicated in several ways. Depending on how it is communicated, the definition of the comparison group can change. Specifically, a comparison group can be defined as a control group, a removal group, or a reversal group. For example, a lift group could receive an SL effect manipulation by being exposed to a positive stereotype. A control, removal, or reversal condition would then be used to evaluate the impact of that lift manipulation. A control condition involves administering either no stereotype

information or irrelevant information to participants (e.g., asking participants to complete a test). A removal condition involves denouncing a stereotype (e.g., stating that men and women perform equally well on a given math test). A reversal condition involves reversing the direction of a stereotype (e.g., stating that women perform better than men on a given math test). Walton and Cohen (2003) previously examined only studies that included removal conditions, but control and reversal conditions also merit consideration because their stereotype saturation might illustrate a degradation in SL effects. Specifically, control, removal, and reversal groups experience different testing instructions that might differently signal their social identities (Turner & Reynolds, 1987). Reversal conditions might reverse the effects of SL, and control conditions' lack of stereotype information might result in weaker effects than removal conditions.

Research Question 2: What are the magnitudes of SL effects for control groups, removal groups, and reversal groups?

Moderator 2: Stereotype Lift Type

Research studies on SL tend to focus on sex and on race. SL studies based on sex often examine the stereotype that men are better than women at math, and SL studies based on race often examine the stereotype of Whites performing better than Blacks on cognitive tests. We do not have a theoretical reason to posit different SL effects for sex than for race. In fact, in the ST domain, the Shewach et al. (2019) meta-analysis reports comparable effects for sex and for race. But since this comparison has been examined in the ST domain, we report SL effects separately for sex and for race to parallel work examining ST.

Research Question 3: What are the magnitudes of sex-based and race-based SL effects?

Moderator 3: Explicitness

Stereotypes can be communicated with varying degrees of explicitness. Nguyen and Ryan (2008) categorized stereotype salient information as blatant, moderately explicit, or subtle/indirect. Blatant communication involves directly telling participants about a stereotype (e.g., stating that men outperform women on this math test). Moderately explicit communication involves stating that there are subgroup performance differences but not stating which subgroup outperforms the other (e.g., stating that there have been performance differences on this math test between men and women). Subtle/indirect communication involves priming participants for a stereotype without direct communication (e.g., having participants indicate their sex before completing a math test). Operational settings should not blatantly communicate stereotypes due to ethical concerns (Ryan & Sackett, 2013), but they are more likely to communicate stereotypes subtly/indirectly. For example, almost all job applications involve initial questions asking for personal history and demographic information. Filling out this information could influence future performance on employee selection tests (Ameri et al., 2018; Kirnan et al., 2009). And more blatant communication of stereotypes might result in a stronger SL effect because they more strongly signal social identities (Turner & Reynolds, 1987).

Research Question 4: What are the magnitudes of SL effects when stereotyped information is communicated blatantly, moderately explicitly, and subtly/indirectly?

Moderator 4: Rewards

SL is a motivational effect (Bandura, 1986; Walton & Cohen, 2003). Nonstereotyped or positively stereotyped group members enjoy an upward social comparison (Cohen & Garcia, 2008), which motivates them to perform better on a test. It is also possible for other forms of motivation to contribute to this performance increase, such as different types of reward systems. SL studies occasionally try to motivate participants through performance-contingent rewards, such as by offering financial bonuses to participants who score well on a cognitive test. This contingent reward could motivate both test takers in SL and non-SL conditions, potentially weakening the SL effect. Note that a finding of SL requires that test takers are not maximizing their effort in the absence of a SL manipulation; receipt of the manipulation is posited to lead to increased effort and thus higher test performance. The presence of contingent rewards based on test performance may serve as an alternate mechanism for motivating test takers to maximize effort. As SL effects are contingent on the potential for test takers to increase their effort in response to the manipulation, SL effects may not be observed in settings in which other mechanisms elicit high effort.

Research Question 5: What are the magnitudes of SL effects in studies not offering contingent rewards and studies offering contingent rewards?

Moderator 5: Research Setting

Almost all SL research is laboratory-based. In such settings, various testing conditions can be examined, such as threat removal and threat reversal conditions. As noted earlier, it is unlikely for such testing instructions to be used in operational testing settings; operational testing instructions usually make no mention of subgroups, much less about subgroups' differential performance on the test. Additionally, laboratory studies can communicate blatant threat manipulations (e.g., "The test you are about to take is one on which members of group X perform better, on average, than members of group Y.") that cannot be done in real testing settings. Shewach et al. (2019) found a noticeably weaker mean ST effect in studies with characteristics that would not be used in operational settings than in studies with features feasible in operational settings. The same effect may be mirrored with SL.

Research Question 6: What are the magnitudes of SL effects in studies with features feasible in operational settings (i.e., using a control group as the comparison group and using subtle/nonexplicit threat manipulations)?

Moderator 6: Publication Status

It is useful to compare published and unpublished findings, as null findings may be less likely to be accepted for publication.

Research Question 7: Are the SL effects comparable in published and unpublished studies?

Publication Bias Analyses

In the ST domain, [Flore and Wicherts \(2015\)](#) found that the overall ST magnitude decreased noticeably after adjusting for publication bias. [Shewach et al. \(2019\)](#) found similar evidence of publication bias in their meta-analysis on ST in adult samples. Despite this publication bias work on ST, there is less information on possible publication bias for SL effects. [Walton and Cohen's \(2003\)](#) meta-analysis of SL included a footnote mentioning that publication status did not predict SL effects with a confidence interval excluding zero. Given the recommendation that modern meta-analyses should ideally employ multiple methods for detecting publication bias ([Inzlicht et al., 2015](#)), we provide a more thorough examination of publication bias in this study (i.e., moderator analysis, trim-and-fill analyses, Egger's test, and the test of excess significance; [Duval & Tweedie, 2000](#); [Egger et al., 1997](#); [Ioannidis & Trikalinos, 2007](#)).

Research Question 8: How does publication bias affect the magnitudes of SL effects?

Method

Transparency and Openness

In the following sections, we describe our sampling plan, all data exclusions, all manipulations, all measures, and adhere to the *Journal of Applied Psychology* methodological checklist. The full data set and the code for analyses have been made publicly available through the Open Science Framework and can be accessed at <https://osf.io/fawmy/>. This research was not preregistered.

Literature Search

We searched relevant literature using multiple strategies, as illustrated in our Preferred Reporting Items for Systematic Reviews and Meta-Analyses diagram ([Page et al., 2021](#); [Appendix](#)). First, we examined the 181 studies cited by [Shewach et al. \(2019\)](#). We then conducted electronic literature searches on Google Scholar for articles that cited [Steele and Aronson \(1995\)](#) or included the terms “stereotype threat,” “stereotype lift,” and “threat removal” from 2001 to 2020, resulting in 918 examined studies. We chose this timeline to reflect when the term “SL” was introduced to the research literature (i.e., early 2000s), most notably by the [Walton and Cohen's \(2003\)](#) meta-analysis. These combined efforts yielded 1,099 potentially eligible studies.

We also searched the gray literature by emailing all lead authors from studies in our final database, asking if they could provide additional unpublished work. We identified more authors to contact by reviewing online conference programs from the Society for Industrial-Organizational Psychology, the Society for Personality and Social Psychology, and the Society for Experimental Social Psychology for the years 2003–2023. One author emailed us a study, but this study did not meet our inclusion criteria.

Criteria for Inclusion

Studies were included in the meta-analysis if they satisfied five criteria. First, the study needed to be experimental with an adult sample (i.e., above 18 years old). Second, the dependent variable needed to be a cognitive measure (e.g., GRE, Scholastic Aptitude

Test, logical reasoning). Third, the study needed to reference an existing stereotype and not a stereotype that was manufactured for the sake of the experiment. An example of manufactured stereotypes is that right-handed people are more intelligent than left-handed people. Fourth, the study needed to include a lift group (i.e., a group for whom a given negative stereotype is not applicable). Finally, enough data had to be reported to permit the calculation of a standardized mean difference (i.e., Cohen's *d*) between the groups. Example statistics include *N*s, means and *SD*s of each group, *F* statistics, and *t* statistics. The study identification and evaluation process are summarized in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses diagram ([Page et al., 2021](#); [Appendix](#)). We collected studies that included only SL effects in addition to studies that included both SL and ST effects.

From the 181 studies cited by [Shewach et al. \(2019\)](#), 23 studies with 31 samples fit our inclusion criteria. Of the 918 studies identified in the electronic search, some were excluded because they were either retracted (2) or not in English (80). Among the remaining 836 studies, only a few (18) were not retrievable, allowing us to assess 818 studies for eligibility. From these possible studies, some were excluded for not including cognitive measures (236), not being empirical (202), not sampling adults (128), and so forth. Our inclusion criteria left us with 22 new eligible studies that had 35 samples. Pairing these with studies identified by [Shewach et al. \(2019\)](#), a total of 45 studies containing 62 independent samples and 5,302 participants were included in the analyses. Seventeen samples provided multiple effect sizes using the same participants (e.g., multiple measurements completed by the same participants). Given that including multiple effect sizes from the same sample violates the assumption of independence ([Hunter & Schmidt, 2004](#)), dependent samples were aggregated. Four studies compared a lift group to multiple different comparison groups (e.g., a control group and a threat removal group). These comparison groups were aggregated to a single effect size for the overall analysis but compared separately with the lift group in moderator analyses, resulting in 62 independent samples for the overall analysis and 66 samples for the comparison group type moderator.

Coding of Study Characteristics and Moderators

The first three authors conducted effect-size coding. These authors first received training from the fourth author and coded sample articles to reach a shared understanding of coding criteria. Two coders separately read and coded a given study. The average interrater agreement was 93.1%. All disagreements were discussed until a consensus was reached. Studies were coded for comparison sample type, lift group, threat group, demographics, manipulation description, relevant statistics to calculate *d*, and moderators included. If information could not be extracted from an article, the article's authors were emailed.

Comparison Group Type. To examine whether effect sizes differ by comparison groups, the type of comparison group used was coded. There were three different comparison groups: control (i.e., no relevant information about subgroup performance), removal (i.e., a statement that subgroups do not differ in performance), and reversal (i.e., a statement that the negatively stereotyped group in fact performs better than the nonstereotyped group). The control groups were given a code of 1, the removal groups were given a code of 2, and the reversal groups were given a code of 3.

Lift Group Type. ST/SL studies differed by the specific stereotype they examined. Thus, the specific group lifted by the stereotype was included as a moderator: race-focused studies were coded as 1, and sex-focused studies were coded as 2. Comparisons across multiple groups were included, such as Whites compared to Blacks or Hispanics. Race- and sex-based comparisons are inherently between-group comparisons, so studies are needed to compare a nonstigmatized group to a stigmatized group. Eight samples did not feature a race- or sex-based comparison and were thus not included in this moderator. Ten samples did not feature comparisons with a nonstigmatized group in the lift condition and a stigmatized group in the threat condition (e.g., comparing lifted women to threatened men on a math test across conditions) and were subsequently removed from this moderator's analyses for more consistent interpretations.

Manipulation Explicitness. Studies can communicate stereotypes blatantly (i.e., directly stating the stereotype), moderately explicitly (i.e., stating that subgroups differ in performance on the test), or subtly/indirectly (i.e., priming participants of their membership in a stereotyped group). Studies that gave blatant ST/SL manipulation were coded as 1, those with a moderately explicit manipulation were coded as 2, and studies with an indirect or subtle manipulation were coded as 3.

Rewards. Studies with no rewards that were contingent on test performance were coded as 0, and those offering any form of contingent rewards were coded as 1. Providing participants with additional financial payment for high cognitive test performance is an example of the reward-contingent condition.

Use of Features Feasible in Operational Settings. Features feasible in operational settings were defined as using a control group and subtle/indirect communication of stereotyped information. Studies with nonoperational setting characteristics were coded as 0, and studies with operational setting characteristics were coded as 1.

Publication Status. Unpublished studies were coded as 0, and published studies were coded as 1.

Meta-Analytic Procedure

The data were analyzed using the "psychmeta" R package (Dahlke & Wiernik, 2019) following Hunter and Schmidt's (2004) psychometric meta-analytic procedures. While the psychometric meta-analysis procedure allows for the correction of sampling and measurement error, we only corrected the former as only five independent samples in our database provided reliability estimates for their dependent measure. The standardized mean difference (d) between different groups was used as the effect size measure for this meta-analysis.

Finally, publication bias of the current data set was evaluated using the trim-and-fill method (Duval & Tweedie, 2000), Egger's test (Egger et al., 1997), and exploratory tests for excess significance (Ioannidis & Trikalinos, 2007). The "metafor" package in R (Viechtbauer, 2010) was utilized to conduct all the publication bias analyses. First, we conducted the trim-and-fill analysis and constructed funnel plots with study sample sizes on the y-axis. This method removes and fills studies to help funnel plots achieve symmetry, recovering the effect estimate and distribution once publication bias is considered. We then employed Egger's test (Egger et al., 1997) to further assess the symmetry of the funnel plots. Finally, we also used the test of excess significance (Ioannidis &

Trikalinos, 2007) to examine whether the observed number of effect sizes with confidence excluding zero is larger than expected.

Results

Table 1 presents the overall meta-analytic estimates and moderator analyses of the effects of SL on cognitive test performance within an adult population. For this table, a negative d value indicates that the lift group performed worse than the comparison group; a positive d value indicates that the lift group performed better than the comparison group.

To answer the first research question about the overall SL effect, the overall mean difference between the SL condition and the comparison condition was $d = 0.09$ ($k = 62$, $N = 5,302$, $SD_{res} = 0.19$). The 95% confidence interval for the overall estimate of the SL effect [0.02, 0.17] did not include zero.

Comparison Group Type

Research Question 2 explored the comparison group type. The mean SL effect was the strongest when compared to a stereotype removal group ($d = 0.14$, $k = 19$, $N = 941$, $SD_{res} = 0.12$), followed by a control group ($d = 0.09$, $k = 44$, $N = 4,476$, $SD_{res} = 0.21$), and the weakest when compared to a reversal group ($d = -0.05$, $k = 3$, $N = 163$, $SD_{res} = 0.00$). The reversal comparison group moderator should be interpreted cautiously, though, due to its comparatively low N and k . Confidence intervals for all these moderators included zero, and their confidence intervals all overlapped.

Lift Group Type

Regarding Research Question 3, the mean SL effect was similar across race ($d = 0.06$, $k = 10$, $N = 1,387$, $SD_{res} = 0.30$) and sex ($d = 0.10$, $k = 34$, $N = 2,733$, $SD_{res} = 0.12$), and their confidence intervals overlapped. It is important to note that there were about three times as many sex-based independent samples than race-based independent samples included in the analyses. Only the sex moderator had a confidence interval that excluded zero.

Explicitness

Research Question 4 addressed how explicitly stereotypes were communicated. The mean SL effect was found to be the largest for studies that included a blatant manipulation ($d = 0.16$, $k = 25$, $N = 1,562$, $SD_{res} = 0.13$) as compared to studies that included either a moderately explicit ($d = 0.07$, $k = 14$, $N = 829$, $SD_{res} = 0.01$) or subtle/indirect manipulations ($d = 0.06$, $k = 23$, $N = 2,911$, $SD_{res} = 0.24$). Only the blatant communication condition yielded a confidence interval that excluded zero, and none of the conditions differed from one another, as evidenced by the overlapping confidence intervals.

Rewards

Research Question 5 investigated the impact of contingent rewards. The mean SL effect was nearly indistinguishable between studies offering no contingent rewards ($d = 0.09$, $k = 56$, $N = 4,958$, $SD_{res} = 0.20$) versus studies offering contingent rewards ($d = 0.10$, $k = 6$, $N = 344$, $SD_{res} = 0.00$). The results should be interpreted

Table 1*Meta-Analyses of Subgroup Differences Between Lift Group and Comparison Group for Study 1*

Category of analysis	<i>k</i>	<i>N</i>	Mean <i>d</i>	<i>SD_d</i>	<i>SD_{res}</i>	95% confidence interval	80% credibility interval
Overall effect size	62	5,302	0.09	0.29	0.19	[0.02, 0.17]	[−0.15, 0.34]
Moderator analyses							
Comparison group type							
Control	44	4,476	0.09	0.29	0.21	[−0.00, 0.17]	[−0.18, 0.35]
Removal	19	941	0.14	0.32	0.12	[−0.02, 0.29]	[−0.02, 0.29]
Reversal	3	163	−0.05	0.29	0.00	[−0.78, 0.67]	[−0.05, −0.05]
Lift group type							
Race	10	1,387	0.06	0.35	0.30	[−0.19, 0.31]	[−0.36, 0.48]
Sex	34	2,733	0.10	0.26	0.12	[0.01, 0.19]	[−0.05, 0.26]
Explicitness							
Blatant	25	1,562	0.16	0.29	0.13	[0.04, 0.28]	[−0.01, 0.33]
Moderately explicit	14	829	0.07	0.27	0.01	[−0.09, 0.23]	[0.05, 0.08]
Indirect and subtle	23	2,911	0.06	0.30	0.24	[−0.07, 0.19]	[−0.26, 0.38]
Rewards							
No contingent rewards	56	4,958	0.09	0.30	0.20	[0.01, 0.17]	[−0.17, 0.36]
Contingent rewards	6	344	0.10	0.20	0.00	[−0.12, 0.31]	[0.10, 0.10]
Setting							
Not operationally feasible	43	2,525	0.13	0.28	0.07	[0.05, 0.22]	[0.05, 0.22]
Operationally feasible	19	2,777	0.06	0.31	0.26	[−0.09, 0.21]	[−0.29, 0.40]
Publication status							
Not published	11	1,175	−0.05	0.37	0.31	[−0.30, 0.19]	[−0.48, 0.37]
Published	51	4,127	0.14	0.26	0.12	[0.06, 0.21]	[−0.02, 0.29]

Note. *k* = number of effect sizes; Mean *d* = mean observed effect sizes; *SD_d* = observed standard deviation; *SD_{res}* = residual standard deviation.

with caution, though, given the small number of studies included in the latter category (*k* = 6). Only the no contingent rewards moderator had a confidence interval that excluded zero, and both effects' confidence intervals overlapped.

Research Setting

Research Question 6 considered the potential role of the research setting. Studies using features not feasible in operational settings produced a stronger mean SL effect (*d* = 0.13, *k* = 43, *N* = 2,525, *SD_{res}* = 0.07) than studies using features feasible in operational settings (*d* = 0.06, *k* = 19, *N* = 2,777, *SD_{res}* = 0.26); however, their confidence intervals overlapped. Only the nonoperational settings moderator yielded confidence intervals that excluded zero.

Publication Status

For Research Question 6, published studies showed larger mean differences (*d* = 0.14, *k* = 51, *N* = 4,127, *SD_{res}* = 0.12) than unpublished studies (*d* = −0.05, *k* = 11, *N* = 1,175, *SD_{res}* = 0.31) for analyses of the SL effect. Only the published studies yielded a confidence interval that excluded zero. The means in the two conditions were not different from one another due to the overlapped confidence intervals.

Publication Bias Analyses

Publication bias has been a common threat to the validity of the conclusions of meta-analyses and systematic reviews (Rothstein, 2008). Results would be biased if there were a persistent and systematic suppression of null findings. To examine the degree of our results being affected by publication bias, we conducted

trim-and-fill analyses (Duval & Tweedie, 2000), Egger's tests (Egger et al., 1997), and the test of excess significance (Ioannidis & Trikalinos, 2007).

We first conducted the test of excess significance, which compares the number of effect size measures that have confidence intervals excluding zero with the number expected given the mean sample size (Ioannidis & Trikalinos, 2007). A much larger number of such findings compared to the number that is expected would serve as evidence of publication bias. The results in Table 2 suggest that there were minimal differences between the number of expected and observed nonzero findings for studies reporting a SL effect, and thus the suppression of studies with confidence intervals including zero due to small sample sizes is not a serious threat to the findings of Study 1.

We additionally conducted trim-and-fill analyses (Duval & Tweedie, 2000) with funnel plots presented in Figure 2. The plot for race studies seemed to be fairly symmetrical, whereas the sex study funnel plot appeared to be rather asymmetrical, indicating a moderate level of publication bias. To further assess the symmetry of the funnel plots, we conducted Egger's test of the intercept (Egger et al., 1997), which employs regression analyses to predict effect sizes using standard error, and a nonzero intercept suggests the presence of publication bias. Results from Egger's test indicate a symmetrical funnel plot for the race group (*t* = −0.10, *p* = .92) yet an asymmetrical plot for the sex group (*t* = 2.83, *p* < .01), suggesting the existence of publication bias for the latter group. The magnitude of the SL effect for race groups reduced (from *d* = 0.12 to *d* = 0.08) after trim and fill. Additionally, the magnitude of the SL effect for sex groups reduced (from *d* = 0.08 to *d* = 0.02) after trim and fill. Therefore, after recomputing the effect sizes based on a symmetrical funnel plot, the SL effect for both race- and sex-based comparisons became smaller, with a more substantial reduction for sex-based comparisons.

Table 2*Test of Excess Significance (Ioannidis & Trikalinos, 2007) for Study 1*

Study	Total <i>k</i>	Expected effects with confidence interval excluding zero ^a	Observed effects with confidence interval excluding zero	$\chi^2(df)$	<i>p</i>
SL effect (Groups A–B)					
Race	10	2.44	2	0.10	.65
Sex	34	3.64	4	0.04	.42

Note. *df* = degrees of freedom; SL = stereotype lift. Expected effects with confidence interval excluding zero calculated based on all samples' cumulative power.

Discussion

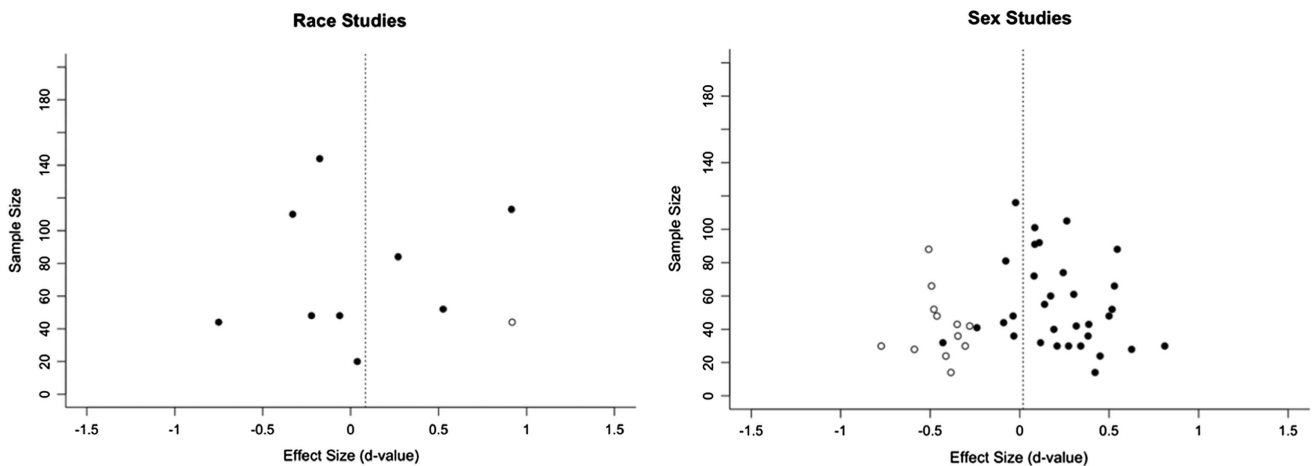
Overall, the findings suggest that the presence of stereotype-inducing information may not have as large of an impact on cognitive test scores as previously reported, as we found a weak mean SL effect ($d = 0.09$, $SD_{res} = 0.19$). While smaller than the Walton and Cohen estimate of $d = 0.24$, our analysis differs from theirs in several ways, notably the focus on only adult samples, the inclusion of comparisons of SL conditions with control and threat reversal conditions in addition to the threat removal conditions examined by Walton and Cohen, and our restriction to studies computing effect sizes based on unadjusted *SDs*. We additionally did not find a larger mean effect ($d = 0.06$, $SD_{res} = 0.26$) when restricting our analysis to studies with features feasible in operational settings. Its confidence interval also includes zero, so we cannot conclude that group differences are further exacerbated by SL effects in operational settings.

The moderator analyses yielded little clear-cut evidence of differences in SL effects on cognitive test performance. Each moderator had fairly large, overlapping confidence intervals, thus making it difficult to interpret the distinct effect of each moderator. We suggest interpreting these moderator analyses with caution due to the small *ks* in many moderator categories.

Given the weak SL effects, it is possible that SL and ST may have a small role in explaining subgroup cognitive test differences. To test this possibility, Study 2 jointly examines within-group SL and ST effects as well as between-group (e.g., race, sex) differences in studies where each of these effects can be assessed, thus permitting head-to-head comparisons of between-group effects and within-group effects.

Study 2: The Joint Effect of SL and ST on Subgroup Mean Differences

Our introduction described how prior work is commonly misinterpreted as indicating that ST fully accounts for subgroup differences on cognitive tests (Sackett et al., 2004; Tomeh & Sackett, 2022). Although that work points out erroneous interpretations, it does not offer a systematic estimate of the joint effect of ST and SL on subgroup mean test differences. To clearly examine the effects of SL and ST on subgroup differences, we focus on a set of studies in which both ST and SL are manipulated. We examine between-group majority–minority mean differences both in the presence and absence of manipulated ST and SL effects. We do so by making comparisons across four groups, using the framework presented in the introduction and summarized in Figure 1.

Figure 2*Funnel Plots for the Trim-and-Fill Analysis Between Lift Group and Lift Comparison Group for Study 1*

Note. Sample-size funnel plots of effect sizes with imputed studies using trim-and-fill method for race (left) and sex studies (right) of Group C versus Group D. The vertical reference lines center on the mean effect size calculated with filled effect sizes.

For each study, we can compute the between-group mean differences in both nonthreat conditions (Groups B and D) to assess subgroup differences in conditions in which neither ST nor SL effects are expected, and the between-group mean differences in both threat conditions (Groups A and C) to assess subgroup differences when both SL and ST effects are present. If large subgroup differences remain in the between-group comparisons in conditions where SL and ST effects are not expected, then previous claims that ST effects fully account for subgroup differences would be challenged.

Our data consist of 32 samples (of the 62 examined in Study 1), with 2,185 participants, that include data for the comparisons outlined in the previous paragraph. We calculate an overall effect size estimate and the moderator analyses from Study 1 for each group comparison. We focus on White–Black and male–female differences since the majority of the stereotype lift literature considers these differences.

Results

We first compare within-group SL and ST effects in our data set with effects in our Study 1 investigation of SL and with Shewach et al.'s (2019) investigation of ST. We do so to evaluate whether our data differ from the databases in the prior work. We obtain an identical mean SL effect of $d = -0.09$ in Study 1 and in the reduced sample we are using in Study 2. Our overall ST mean estimate is $d = -0.18$, which is smaller than the mean $d = -0.31$ obtained by Shewach et al. However, in the reduced set of samples with features feasible in operational settings, we obtain a mean $d = -0.14$, which is identical to that obtained by Shewach et al.

We present two sets of meta-analyses. Table 3 shows the between-group comparison for conditions in which neither SL nor

ST are posited to occur (e.g., the Group B–D comparison). In contrast, Table 4 shows the between-group comparison in SL and ST conditions (e.g., the Group A–C comparison). For these tables, a negative d value indicates that the first group performed worse than the second group; a positive d value indicates that the first group performed better than the second group. For example, a positive d value for our comparison across Groups B–D would indicate that Group B performed better on average than Group D.

Table 3 shows the between-group mean difference in nonthreat/nonlift conditions (e.g., comparing the nonthreatened vs. threatened groups with both in control group conditions). For example, it could compare men and women taking a math test in control conditions. While the table reports an overall mean effect, we do not view it as informative, as the overall analysis combines male–female differences and White–Black differences. These are best interpreted separately, so we attend to the male–female and White–Black differences. The male–female mean difference in nonthreat conditions was $d = 0.23$ ($k = 27$, $N = 2,023$, $SD_{\text{res}} = 0.41$) with a 95% confidence interval of [0.04, 0.41]. The White–Black mean difference in nonthreat conditions was $d = 0.91$ ($k = 5$, $N = 348$, $SD_{\text{res}} = 0.00$) with a 95% confidence interval of [0.64, 1.18]. The table includes additional moderator analyses, but we do not discuss them as the confidence intervals overlap for all moderators.

Table 4 shows the between-group mean difference between the nonthreatened group under a lift manipulation and the threatened group under a threat manipulation condition. For example, it examines conditions like men and women taking a math test after being told stereotype-inducing information that men perform better than women on math tests. Table 4 also reports an overall mean effect that we do not find informative because the overall analysis combines male–female differences and White–Black differences, so we attend to each of the comparisons separately. The male–female

Table 3

Meta-Analyses of Subgroup Differences Between Lift Comparison Group (Group D) and Threat Comparison Group (Group B) for Study 2

Category of analysis	<i>k</i>	<i>N</i>	Mean <i>d</i>	<i>SD_d</i>	<i>SD_{res}</i>	95% confidence interval	80% credibility interval
Overall effect size	32	2,371	0.33	0.51	0.44	[0.14, 0.51]	[−0.26, 0.91]
Moderator analyses							
Comparison group type							
Control	23	1,896	0.34	0.45	0.39	[0.14, 0.53]	[−0.17, 0.85]
Removal	10	589	0.09	0.36	0.24	[−0.17, 0.35]	[−0.24, 0.42]
Reversal	3	197	0.31	0.31	0.17	[−0.46, 1.09]	[−0.02, 0.64]
Lift group type							
Race	5	348	0.91	0.22	0.00	[0.64, 1.18]	[0.91, 0.91]
Sex	27	2,023	0.23	0.47	0.41	[0.04, 0.41]	[−0.31, 0.76]
Explicitness							
Blatant	13	967	0.27	0.47	0.40	[−0.01, 0.56]	[−0.26, 0.81]
Moderately explicit	10	480	0.03	0.61	0.53	[−0.41, 0.46]	[−0.71, 0.76]
Indirect and subtle	9	924	0.54	0.42	0.37	[0.21, 0.86]	[0.02, 1.05]
Rewards							
No contingent rewards	26	1,987	0.24	0.38	0.29	[0.08, 0.39]	[−0.15, 0.62]
Contingent rewards	6	384	0.79	0.81	0.76	[−0.05, 1.64]	[−0.33, 1.92]
Setting							
Not operationally feasible	24	1,483	0.19	0.51	0.44	[−0.03, 0.41]	[−0.39, 0.77]
Operationally feasible	8	888	0.55	0.43	0.37	[0.19, 0.91]	[0.02, 1.08]
Publication status							
Not published	6	777	0.42	0.25	0.17	[0.16, 0.68]	[0.17, 0.67]
Published	26	1,594	0.28	0.60	0.53	[0.04, 0.52]	[−0.41, 0.97]

Note. *k* = number of effect sizes; Mean *d* = mean observed effect sizes; *SD_d* = observed standard deviation; *SD_{res}* = residual standard deviation.

Table 4*Meta-Analyses of Subgroup Differences Between Lift Group (Group C) and Threat Group (Group A) for Study 2*

Category of analysis	<i>k</i>	<i>N</i>	Mean <i>d</i>	<i>SD_d</i>	<i>SD_{res}</i>	95% confidence interval	80% credibility interval
Overall effect size	32	2,428	0.49	0.41	0.32	[0.35, 0.64]	[0.07, 0.91]
Moderator analyses							
Comparison group type							
Control	23	1,927	0.48	0.42	0.35	[0.29, 0.66]	[0.02, 0.93]
Removal	10	614	0.53	0.35	0.23	[0.28, 0.78]	[0.21, 0.85]
Reversal	3	208	0.47	0.12	0.00	[0.17, 0.77]	[0.47, 0.47]
Lift group type							
Race	5	351	1.02	0.35	0.18	[0.59, 1.45]	[0.74, 1.29]
Sex	27	2,077	0.41	0.35	0.25	[0.27, 0.54]	[0.08, 0.73]
Explicitness							
Blatant	13	944	0.53	0.34	0.23	[0.32, 0.73]	[0.21, 0.84]
Moderately explicit	10	522	0.51	0.27	0.00	[0.31, 0.70]	[0.51, 0.51]
Indirect and subtle	9	962	0.45	0.54	0.50	[0.04, 0.87]	[-0.24, 1.15]
Rewards							
No contingent rewards	26	2,020	0.44	0.38	0.29	[0.28, 0.59]	[0.05, 0.82]
Contingent rewards	6	408	0.77	0.44	0.36	[0.30, 1.24]	[0.24, 1.30]
Setting							
Not operationally feasible	24	1,502	0.51	0.31	0.15	[0.38, 0.65]	[0.32, 0.71]
Operationally feasible	8	926	0.46	0.56	0.51	[-0.00, 0.92]	[-0.26, 1.19]
Publication status							
Not published	6	814	0.26	0.46	0.42	[-0.22, 0.74]	[-0.37, 0.88]
Published	26	1,614	0.61	0.34	0.19	[0.48, 0.75]	[0.36, 0.86]

Note. *k* = number of effect sizes; Mean *d* = mean observed effect sizes; *SD_d* = observed standard deviation; *SD_{res}* = residual standard deviation.

mean difference in the presence of threat and lift manipulations was $d = 0.41$ ($k = 27$, $N = 2,077$, $SD_{res} = 0.25$) with a 95% confidence interval of [0.27, 0.54]. The White-Black mean difference in the presence of threat and lift manipulations was $d = 1.02$ ($k = 5$, $N = 351$, $SD_{res} = 0.18$) with a 95% confidence interval of [0.59, 1.45]. Thus, the male-female differences are larger in the presence of threat and lift manipulation than when those manipulations are not present ($d = 0.41$ vs. $d = 0.23$); this is also the case for the White-Black comparison ($d = 1.02$ vs. $d = 0.91$). The table also includes additional moderator analyses, which we do not discuss because their confidence intervals largely overlap.

We conducted publication bias analyses on these effects through the test of excess significance (Ioannidis & Trikalinos, 2007).

As shown in Table 5, there was evidence for publication bias with the sex-based comparisons in the presence (Groups A-C) and absence of ST and SL (Groups B-D) because the number of observed effects with confidence intervals excluding zero was significantly different from the expected number of such significant effects.

We additionally conducted trim-and-fill analyses (Duval & Tweedie, 2000). This procedure's results are summarized in Table 6. These results are expanded upon by the Egger's test (Egger et al., 1997). Results indicated symmetrical plots for the race group of the Group A-C comparison ($t = -0.82$, $p = .47$) and both the race ($t = 0.52$, $p = .64$) and sex group ($t = -0.75$, $p = .46$) for the Group B-D comparison, yet the test suggested an asymmetrical

Table 5*Test of Excess Significance (Ioannidis & Trikalinos, 2007) for Study 2*

Study	Total <i>k</i>	Expected effects with confidence interval excluding zero ^a	Observed effects with confidence interval excluding zero	$\chi^2(df)$	<i>p</i>
Between-group effect with SL and ST present (Groups A-C)					
Race	5	4.52	4	0.62	.78
Sex	27	10.16	16	5.37*	.01
Between-group effect in nonthreat and nonlift conditions (Groups B-D)					
Race	5	4.40	4	0.29	.71
Sex	27	8.20	7	0.25	.69

Note. *df* = degrees of freedom; SL = stereotype lift; ST = stereotype threat. Expected effects with confidence interval excluding zero calculated based on all samples' cumulative power.

* $p < .05$.

Table 6

Group Differences Before and After Trim-and-Fill Analyses for Study 2

Study	Before	After
Between-group effect with SL and ST present (Groups A–C)		
Race	1.02	1.03
Sex	0.41	0.29
Between-group effect in nonthreat and nonlift conditions (Groups B–D)		
Race	0.91	0.92
Sex	0.23	0.35

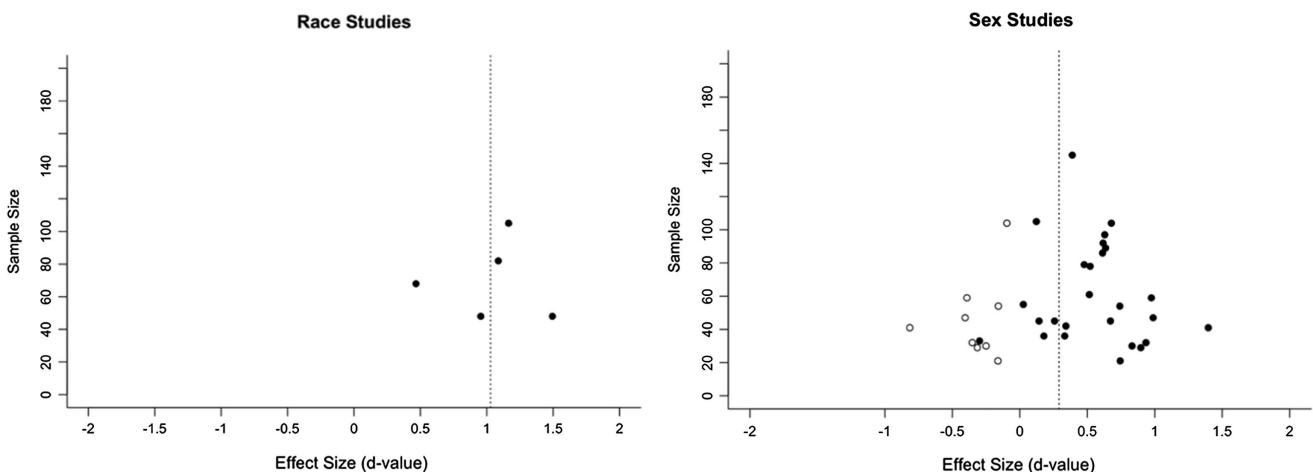
Note. SL = stereotype lift; ST = stereotype threat. Expected effects with confidence interval excluding zero calculated based on all samples' cumulative power.

plot for the sex group of the Group A–C comparison ($t = 3.65$, $p < .01$), indicating the existence of publication bias for that comparison.

For race, SL and ST contributed to White–Black mean differences, as the mean difference in conditions where stereotype-inducing information was present was $d = 1.03$ versus $d = 0.92$ in conditions in which stereotype-inducing information was not present. In other words, the presence of SL and ST increased the standardized performance differences between Whites and Blacks by 0.11. A weaker pattern appeared with sex-based comparisons, as their mean difference in conditions with stereotype-inducing information present was $d = 0.29$ versus $d = 0.35$ in conditions in which stereotype-inducing information was not present. For sex-based comparisons, the presence of SL and ST effects reduced performance differences between men and women by 0.06. Figures 3 and 4 illustrate these performance differences. It should be noted, though, that there were noticeably fewer race-based studies ($k = 5$) than sex-based studies ($k = 27$) to substantiate these conclusions.

Figure 3

Funnel Plots for the Trim-and-Fill Analysis Between the Lift Group (Group C) and Threat Group (Group A) for Study 2



Note. Sample-size funnel plots of effect sizes with imputed studies using trim-and-fill method for race (left) and sex studies (right) of Group C versus Group A. The vertical reference lines center on the mean effect size calculated with filled effect sizes.

Overall Discussion

Stereotype Lift

This study examined the extent to which SL and ST influence subgroup performance differences on cognitive tests. We first presented an updated meta-analysis of SL effects. We found a mean $d = 0.09$ ($SD_{res} = 0.19$) across all studies and a value of $d = 0.06$ ($SD_{res} = 0.26$) in studies with features feasible in operational testing settings. This $d = 0.06$ value in studies with features feasible in operational settings had a confidence interval that did not exclude zero. We therefore conclude that SL effects, if present, are small.

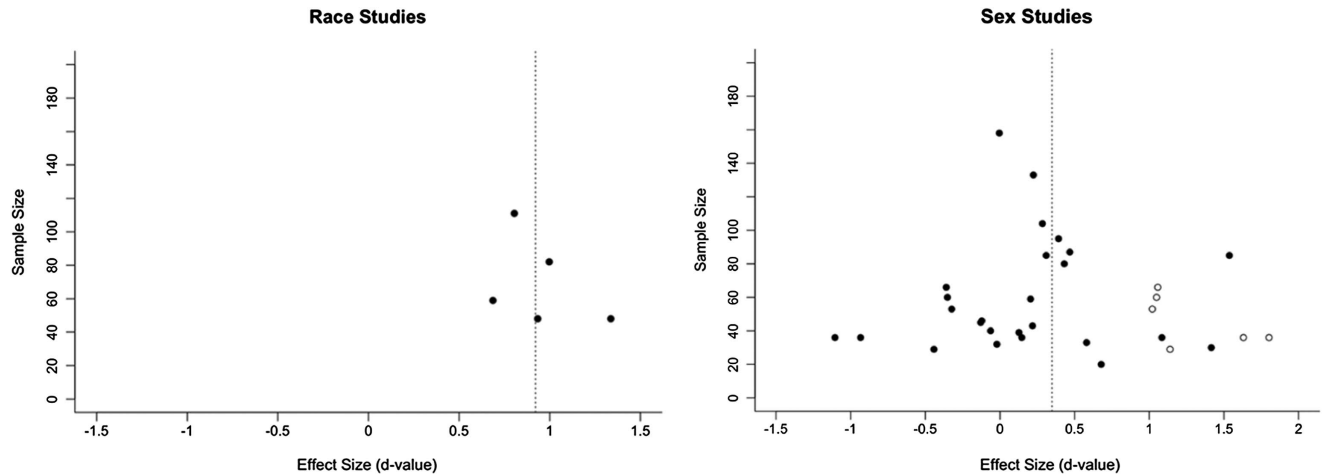
We suggest that SL effects may be specific to the experimental paradigm used. Participants are typically college students participating for course credits. Absent performance incentives, the issue of motivation looms large. Do examinees exert the maximum effort when tested? A finding of SL would indicate they do not: when nonstereotyped test takers are primed about the stereotype disadvantaging another group, they are presumably motivated to focus attentional resources on the test and subsequently perform better. But we do not see this mechanism as likely in operational testing settings because candidates inherently have a strong motivation to exert maximum effort in pursuit of a desired outcome (e.g., a job or admission to an educational program).

Stereotype Lift and Threat Effects on Subgroup Differences

We then turned to a subset of the studies that permitted analyses of the joint effects of SL and ST on male–female and White–Black differences on cognitive tests. We examined these mean differences in conditions in which both SL and ST were imposed by experimental manipulation and in comparison conditions in which neither SL nor ST were imposed. From our meta-analytic results, we found the mean $d = 1.02$ for the White–Black performance differences in the SL/ST conditions compared to the $d = 0.91$ in the

Figure 4

Funnel Plots for the Trim-and-Fill Analysis Between the Lift Comparison Group (Group D) and Threat Comparison Group (Group B) for Study 2



Note. Sample-size funnel plots of effect sizes with imputed studies using trim-and-fill method for race (left) and sex studies (right) of Group D versus Group B. The vertical reference lines center on the mean effect size calculated with filled effect sizes.

no SL/ST comparison group condition. We additionally found the mean $d = 0.41$ for the male–female performance differences in the SL/ST conditions and the $d = 0.23$ in the no SL/ST comparison group condition. Thus, the joint SL/ST effects increase subgroup performance differences on cognitive tests for race- and sex-based comparisons. The race-based findings must be treated cautiously, however, due to the small number of studies used to substantiate these conclusions. We also note that the sex differences between threat and nonthreat conditions were reduced after trim-and-fill analyses, suggesting the existence of publication bias. The fact that both race- and sex-based differences remain substantial in nonthreat conditions challenged claims that ST effects fully account for subgroup differences on cognitive tests (Sackett et al., 2004; Tomeh & Sackett, 2022).

Given the finding that ST and SL effects had relatively small influences on adult cognitive performance, it is important to consider other potential explanations for the subgroup difference values found in the literature. It is beyond the scope of this study to fully review the multitude of potential sociodemographic and systematic factors contributing to subgroup differences in cognitive ability tests; however, we offer several potential contributors that point to the role of generational social inequality and subsequent differential access to resources (Newman et al., 2007). Outtz and Newman (2010) acknowledged the role of socioeconomic status and other environmental characteristics in cognitive test performance. Cottrell et al. (2015) expanded upon Outtz and Newman (2010) to propose a model of adverse impact that suggests that Black–White mean differences in cognitive test scores arise from various racially disparate conditions including, family income and maternal access to education.

Implications for Research and Practice

Our findings regarding SL effects and Shewach et al.'s (2019) findings on ST effects indicate that both effects are quite small in

settings with features that are viable in operational settings. That said, the vast majority of studies are in laboratory settings rather than in actual operational testing settings. Controlled experimental studies in operational settings are rare; the few ST studies that exist in operational settings were reviewed by Sackett and Ryan (2011) and show near-zero effects. Further work in operational settings would be useful, given the imbalance of studies in laboratory versus operational settings.

It is important to note that the finding of small SL and ST effects on test performance should not be interpreted as a blanket rejection of ST and SL as potentially important psychological phenomena. For example, our findings that SL and ST have small effects on test takers' abilities to demonstrate their current standing on the attributes measured by those tests do not speak to other questions, such as whether stereotypes affect the learning of new material. Several experiments in classroom setting suggest that threat reduction efforts can aid subsequent learning (e.g., Cohen et al., 2006, 2009). Research that carefully differentiates between the settings in which stereotypes are examined has the potential to yield useful insights into an understanding of differential stereotype effects across settings.

In terms of implications for practice, the emerging message that SL and ST effects appear small in operational settings implies that those running operational testing programs should not count on substantially reducing majority–minority mean differences on test scores via altering the testing environment. Strategies such as broadening the range of constructs addressed in a selection system are likely to have much larger effects on reducing subgroup mean differences (Ployhart & Holtz, 2008).

Even if stereotype effects are small, efforts with the possibility of alleviating them remain worthwhile. A wide variety of suggestions have been put forward as to ways of reducing ST in a testing environment; Ryan and Sackett (2013) reviewed and commented on which could feasibly be implemented. Feasible suggestions include

emphasizing the malleability of the construct being tested, showing diverse role models in pretest materials, using less “charged” labels (e.g., problem solving vs. intelligence), and providing a positive and encouraging test environment. Some other suggestions involve presenting information that may be factually untrue, such as asserting that the stereotype is incorrect (e.g., no mean differences, where mean differences are, in fact, found) or presenting the test as innocuous (i.e., as low stakes, when it is, in fact, used for selection decisions). Ryan and Sackett argued that one cannot ethically present untrue information to an applicant, thus discrediting these suggestions for operational testing settings.

Limitations

We note that the number of studies included for several moderator categories is small (e.g., $k = 3$ in Study 1 for the “reversal” comparison group condition and $k = 6$ for the “contingent rewards” motivation category). Small k values for these conditions made associated conclusions more difficult to reach and increased the possibility of making a Type-II error.

Furthermore, our sample is overwhelmingly composed of college students. College students are commonly 18–22 years of age and thus at a different stage of cognitive development than older adults. It is possible that college students perform fundamentally differently than older adults on cognitive tests due to their prefrontal cortex’s ongoing development (Arain et al., 2013). The prefrontal cortex is important in executive functions, including working memory and other characteristic skills assessed by cognitive tests (Ones et al., 2017), so sampling bias may have impacted the overall results. A reviewer raised an additional issue about the largely college sample, namely, whether these findings generalize to less educated individuals. We do not have a direct answer, as data on such samples are lacking. However, as we noted earlier, there is extensive research on ST effects with child and adolescent samples. A meta-analysis of that literature by Flore and Wicherts (2015) reports mean ST effects that are comparable in size to those reported in the meta-analysis of largely college samples by Shewach et al. (2019). These findings suggest that the effects are not specific to college samples.

Another limitation within this study concerns the limited set of stereotypes examined. There are many stereotypes that could influence cognitive test scores in operational settings, yet our study largely focused on sex- and race-based stereotypes. These stereotypes fail to represent the full array of possible stereotypes in existence. Other types of cognitive stereotypes include Northerners being more intelligent than Southerners (Clark et al., 2011) and younger people being better than older people at memory tasks (Chasteen et al., 2012; Hess et al., 2003). Because the SL and ST literatures focus consistently on race- and sex-based stereotypes (often measured with mathematics tests), it is still challenging to generalize these findings to different stereotypes, as they may yield fundamentally different effects.

Future Directions

We caution future meta-analyses against including data that are adjusted on a prior covariate (e.g., prior cognitive test performance)

when estimating subgroup differences. Doing so artificially reduces variance in the criterion variable and upwardly biases d values (Shewach et al., 2019). This issue may come into play in other domains as well, as the use of control variables is common in research.

We also believe that future SL and ST studies in general would benefit from reporting unadjusted test performance descriptive statistics, reporting more detailed sample demographics, and conducting more studies in operational settings (rather than just laboratory settings). Hunter and Schmidt (2004) famously stated that “every study should be viewed as a data point to be contributed to a larger meta-analysis” (p. 12). Subsequently, these improvements to the SL literature could help produce more precise meta-analyses in the future.

We note that there are additional potential moderators not explored in the present analyses that could be explored in the future. One is the awareness of the stereotype in question. A reviewer offered the interesting suggestion that members of the nonstereotyped group may be less likely to be aware of the stereotype than members of the stereotyped group. Clearly, both ST and SL effects require awareness of the stereotype. One contributor to smaller SL effects than ST effects could be lower awareness in the nonstereotyped group. Aggregated effects reported in meta-analyses are clearly group averages, and failure to explore this moderator does not bias conclusions about the average effects of ST and SL. Nonetheless, more nuanced findings are possible if samples could be differentiated on the basis of stereotype awareness. A similar issue applies to the concept of group identification, which has long been viewed as a precondition for experiencing ST. Should minority group identification be stronger than majority group identification, this could also contribute to the finding of smaller SL effects than ST effects.

A final moderator suggested by a reviewer is the framing of a stereotype manipulation: Does it matter if the message conveyed is “men perform better than women on average on this test” or “women perform worse than men on average on this test”? An examination of our database reveals that the former (men perform better than women) is so dominant that a comparison is not possible with the data at hand. Given the literature on framing effects (e.g., Rosette & Koval, 2018), this could be explored in future work.

Conclusion

SL and ST effects were found to have a relatively small influence on adult cognitive test performance for both race- and sex-based comparisons. These findings confirmed previous meta-analytic estimates for ST effects but differed from Walton and Cohen’s (2003) meta-analysis on SL, which estimated a larger effect. This difference could result from Walton and Cohen possibly including adjusted descriptive statistics, which would bias their estimates upward. However, given that Walton and Cohen’s study also included child samples, whereas the present study included only adult samples, it is also possible that the SL effect fundamentally differs for children and adults. Furthermore, the overlapping confidence intervals across our moderator analyses suggested that there are few clear-cut moderating effects on the presence of

stereotype-salient information and cognitive test scores. Finally, the meta-analytic finding of White–Black differences and male–female differences in conditions in which ST and SL are not imposed experimentally challenges assertions that stereotypes fully account for subgroup cognitive test performance differences. Further work is needed to determine the causes of these subgroup differences and reduce their effects. Decision makers for the operational use of cognitive tests need to consider issues including validity evidence for the context in question, merits of alternate predictors, and evidence regarding potential predictive bias in order to make informed decisions about test use.

References

References marked with an asterisk indicate studies included in the meta-analyses.

- *Adams, G., Garcia, D. M., Purdie-Vaughns, V., & Steele, C. M. (2006). The detrimental effects of a suggestion of sexism in an instruction situation. *Journal of Experimental Social Psychology*, 42(5), 602–615. <https://doi.org/10.1016/j.jesp.2005.10.004>
- Ameri, M., Schur, L., Adya, M., Bentley, F. S., McKay, P., & Kruse, D. (2018). The disability employment puzzle: A field experiment on employer hiring behavior. *Industrial & Labor Relations Review*, 71(2), 329–364. <https://doi.org/10.1177/0019793917717474>
- *Anderson, R. D. (2001). *Stereotype threat: The effects of gender identification on standardized test performance* (Publication No. 3007927) [Doctoral dissertation, James Madison University]. ProQuest Dissertations and Theses Global.
- Araín, M., Haque, M., Johal, L., Mathur, P., Nel, W., Rais, A., Sandhu, R., & Sharma, S. (2013). Maturation of the adolescent brain. *Neuropsychiatric Disease and Treatment*, 9, 449–461. <https://doi.org/10.2147/ndt.s39776>
- Bandura, A. (1986). The explanatory and predictive scope of self-efficacy theory. *Journal of Social and Clinical Psychology*, 4(3), 359–373. <https://doi.org/10.1521/jscp.1986.4.3.359>
- Beilock, S. L., & McConnell, A. R. (2004). Stereotype threat and sport: Can athletic performance be threatened? *Journal of Sport & Exercise Psychology*, 26(4), 597–609. <https://doi.org/10.1123/jsep.26.4.597>
- *Brown, R. P., & Day, E. A. (2006). The difference isn't black and white: Stereotype threat and the race gap on raven's advanced progressive matrices. *Journal of Applied Psychology*, 91(4), 979–985. <https://doi.org/10.1037/0021-9010.91.4.979>
- *Chaffee, K. E., Lou, N. M., & Noels, K. A. (2020). Does stereotype threat affect men in language domains? *Frontiers in Psychology*, 11, Article 1302. <https://doi.org/10.3389/fpsyg.2020.01302>
- Chasteen, A. L., Kang, S. K., & Remedios, J. D. (2012). Aging and stereotype threat: Development, process, and interventions. In M. Inzlicht & T. Schmader (Eds.), *Stereotype threat: Theory, process, and application* (pp. 202–216). Oxford University Press.
- *Chateignier, C., Dutrévis, M., Nugier, A., & Chekroun, P. (2009). French-Arab students and verbal intellectual performance: Do they really suffer from a negative intellectual stereotype? *European Journal of Psychology of Education*, 24(2), 219–234. <https://doi.org/10.1007/BF03173013>
- Clark, J. K., Eno, C. A., & Guadagno, R. E. (2011). Southern discomfort: The effects of stereotype threat on the intellectual performance of US southerners. *Self and Identity*, 10(2), 248–262. <https://doi.org/10.1080/15298861003771080>
- Cohen, G. L., & Garcia, J. (2008). Identity, belonging, and achievement: A model, interventions, implications. *Current Directions in Psychological Science*, 17(6), 365–369. <https://doi.org/10.1111/j.1467-8721.2008.00607.x>
- Cohen, G. L., Garcia, J., Apfel, N., & Master, A. (2006). Reducing the racial achievement gap: A social-psychological intervention. *Science*, 313(5791), 1307–1310. <https://doi.org/10.1126/science.1128317>
- Cohen, G. L., Garcia, J., Purdie-Vaughns, V., Apfel, N., & Brzustoski, P. (2009). Recursive processes in self-affirmation: Intervening to close the minority achievement gap. *Science*, 324(5925), 400–403. <https://doi.org/10.1126/science.1170769>
- Cottrell, J. M., Newman, D. A., & Roisman, G. I. (2015). Explaining the Black–White gap in cognitive test scores: Toward a theory of adverse impact. *Journal of Applied Psychology*, 100(6), 1713–1736. <https://doi.org/10.1037/apl0000020>
- *Croizet, J. C., Després, G., Gauzins, M. E., Huguet, P., Leyens, J. P., & Méot, A. (2004). Stereotype threat undermines intellectual performance by triggering a disruptive mental load. *Personality and Social Psychology Bulletin*, 30(6), 721–731. <https://doi.org/10.1177/0146167204263961>
- Dahlke, J. A., & Wiernik, B. M. (2019). psychmeta: An R package for psychometric meta-analysis. *Applied Psychological Measurement*, 43(5), 415–416. <https://doi.org/10.1177/0146621618795933>
- *Danso, H. A., & Esses, V. M. (2001). Black experimenters and the intellectual test performance of White participants: The tables are turned. *Journal of Experimental Social Psychology*, 37(2), 158–165. <https://doi.org/10.1006/jesp.2000.1444>
- Duval, S., & Tweedie, R. (2000). Trim and fill: A simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics*, 56(2), 455–463. <https://doi.org/10.1111/j.0006-341X.2000.00455.x>
- Egger, M., Davey Smith, G., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *The BMJ*, 315(7109), Article 629. <https://doi.org/10.1136/bmj.315.7109.629>
- *Elizaga, R. A., & Markman, K. D. (2008). Peers and performance: How in-group and out-group comparisons moderate stereotype threat effects. *Current Psychology*, 27(4), 290–300. <https://doi.org/10.1007/s12144-008-9041-y>
- *Eriksson, K., & Lindholm, T. (2007). Making gender matter: The role of gender-based expectancies and gender identification on women's and men's math performance in Sweden. *Scandinavian Journal of Psychology*, 48(4), 329–338. <https://doi.org/10.1111/j.1467-9450.2007.00588.x>
- Flore, P. C., & Wicherts, J. M. (2015). Does stereotype threat influence performance of girls in stereotyped domains? A meta-analysis. *Journal of School Psychology*, 53(1), 25–44. <https://doi.org/10.1016/j.jsp.2014.10.002>
- *Fogliati, V. J., & Bussey, K. (2013). Stereotype threat reduces motivation to improve: Effects of stereotype threat and feedback on women's intentions to improve mathematical ability. *Psychology of Women Quarterly*, 37(3), 310–324. <https://doi.org/10.1177/0361684313480045>
- *Good, C., Aronson, J., & Harder, J. A. (2008). Problems in the pipeline: Stereotype threat and women's achievement in high-level math courses. *Journal of Applied Developmental Psychology*, 29(1), 17–28. <https://doi.org/10.1016/j.appdev.2007.10.004>
- *Grand, J. A., Ryan, A. M., Schmitt, N., & Hmurovic, J. (2010). How far does stereotype threat reach? The potential detriment of face validity in cognitive ability testing. *Human Performance*, 24(1), 1–28. <https://doi.org/10.1080/08959285.2010.518184>
- Hess, T. M., Auman, C., Colcombe, S. J., & Rahhal, T. A. (2003). The impact of stereotype threat on age differences in memory performance. *The Journals of Gerontology: Series B*, 58(1), 3–11. <https://doi.org/10.1093/geronb/58.1.P3>
- *Hirnstien, M., Freund, N., & Hausmann, M. (2012). Gender stereotyping enhances verbal fluency performance in men (and women). *Zeitschrift für Psychologie*, 220(2), 70–77. <https://doi.org/10.1027/2151-2604/a000098>
- Hunter, J. E., & Schmidt, F. L. (2004). *Methods of meta-analysis: Correcting error and bias in research findings*. Sage Publications.

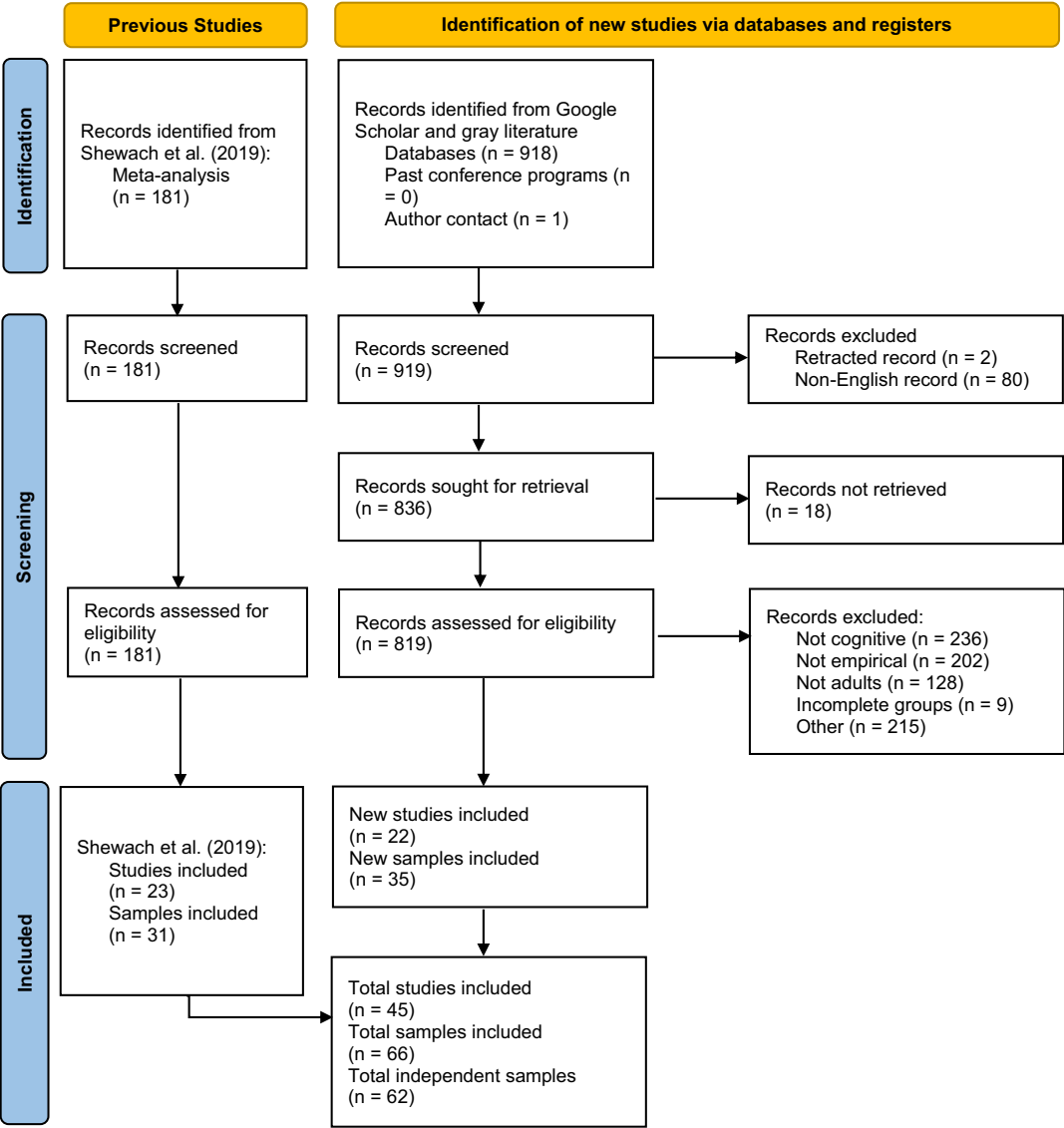
- *Ihme, T. A., & Möller, J. (2015). "He who can, does; he who cannot, teaches?": Stereotype threat and preservice teachers. *Journal of Educational Psychology*, 107(1), 300–308. <https://doi.org/10.1037/a0037373>
- *Inzlicht, M. (2001). *A threatening intellectual environment: When and why females are susceptible to experiencing problem-solving deficits in the presence of males* [Doctoral dissertation, Brown University]. Academia.
- Inzlicht, M., Gervias, W., & Berkman, E. (2015). *Bias-correction techniques alone cannot determine whether ego depletion is different from zero: Commentary on Carter, Kofler, Forster, & McCullough, 2015* [Unpublished manuscript]. <https://doi.org/10.2139/ssm.2659409>
- Ioannidis, J. P., & Trikalinos, T. A. (2007). An exploratory test for an excess of significant findings. *Clinical Trials*, 4(3), 245–253. <https://doi.org/10.1177/1740774507079441>
- *Jamieson, J. P., & Harkins, S. G. (2009). The effect of stereotype threat on the solving of quantitative GRE problems: A mere effort interpretation. *Personality and Social Psychology Bulletin*, 35(10), 1301–1314. <https://doi.org/10.1177/0146167209335165>
- *Johns, M., Inzlicht, M., & Schmader, T. (2008). Stereotype threat and executive resource depletion: Examining the influence of emotion regulation. *Journal of Experimental Psychology: General*, 137(4), 691–705. <https://doi.org/10.1037/a0013834>
- *Johnson, H. J., Barnard-Brak, L., Saxon, T. F., & Johnson, M. K. (2012). An experimental study of the effects of stereotype threat and stereotype lift on men and women's performance in mathematics. *Journal of Experimental Education*, 80(2), 137–149. <https://doi.org/10.1080/00220973.2011.567312>
- *Jones, P. R. (2011). Reducing the impact of stereotype threat on women's math performance: Are two strategies better than one? *Electronic Journal of Research in Educational Psychology*, 9(2), 587–616. <https://doi.org/10.25115/ejrep.v9i24.1458>
- *Kirman, J. P., Alfieri, J. A., Bragger, J. D., & Harris, R. S. (2009). An investigation of stereotype threat in employment tests. *Journal of Applied Social Psychology*, 39(2), 359–388. <https://doi.org/10.1111/j.1559-1816.2008.00442.x>
- *McGlone, M. S., & Aronson, J. (2006). Stereotype threat, identity salience, and spatial reasoning. *Journal of Applied Developmental Psychology*, 27(5), 486–493. <https://doi.org/10.1016/j.appdev.2006.06.003>
- *McKay, P. F. (1999). *Stereotype threat and its effect on the cognitive ability test performance of African-Americans: The development of a theoretical model* (Publication No. 9925150) [Doctoral dissertation, University of Akron]. ProQuest Dissertations and Theses Global.
- Newman, D. A., Hanges, P. J., & Outtz, J. L. (2007). Racial groups and test fairness, considering history and construct validity. *American Psychologist*, 62(9), 1082–1083. <https://doi.org/10.1037/0003-066X.62.9.1082>
- *Nguyen, H. H. D., O'Neal, A., & Ryan, A. M. (2003). Relating test-taking attitudes and skills and stereotype threat effects to the racial gap in cognitive ability test performance. *Human Performance*, 16(3), 261–293. https://doi.org/10.1207/S15327043HUP1603_5
- Nguyen, H. H. D., & Ryan, A. M. (2008). Does stereotype threat affect test performance of minorities and women? A meta-analysis of experimental evidence. *Journal of Applied Psychology*, 93(6), 1314–1334. <https://doi.org/10.1037/a0012702>
- *O'Brien, L. T., & Hummert, M. L. (2006). Memory performance of late middle-aged adults: Contrasting self-stereotyping and stereotype threat accounts of assimilation to age stereotypes. *Social Cognition*, 24(3), 338–358. <https://doi.org/10.1521/soco.2006.24.3.338>
- Ones, D. S., Viswesvaran, C., & Dilchert, S. (2017). Cognitive ability in personnel selection decisions. In A. Evers, N. Anderson, & O. Voskuil (Eds.), *The Blackwell handbook of personnel selection* (pp. 143–173). Wiley. <https://doi.org/10.1002/9781405164221.ch7>
- Outtz, J. L., & Newman, D. A. (2010). A theory of adverse impact. In J. L. Outtz (Ed.), *Adverse impact: Implications for organizational staffing and high stakes selection* (pp. 53–94). Routledge. <https://doi.org/10.4324/9780203848418>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., ... Moher, D. (2021). PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *BMJ*, 372. <https://doi.org/10.1136/bmj.n160>
- Ployhart, R. E., & Holtz, B. C. (2008). The diversity–validity dilemma: Strategies for reducing racioethnic and sex subgroup differences and adverse impact in selection. *Personnel Psychology*, 61(1), 153–172. <https://doi.org/10.1111/j.1744-6570.2008.00109.x>
- *Ployhart, R. E., Ziegert, J. C., & McFarland, L. A. (2003). Understanding racial differences on cognitive ability tests in selection contexts: An integration of stereotype threat and applicant reactions research. *Human Performance*, 16(3), 231–259. https://doi.org/10.1207/S15327043HUP1603_4
- *Rahe, M., Ruthsatz, V., Jansen, P., & Quaiser-Pohl, C. (2018). Influence of sex-stereotyped stimuli on the mental-rotation performance of elderly persons. *Experimental Aging Research*, 44(4), 284–296. <https://doi.org/10.1080/0361073X.2018.1475156>
- *Rea, J. N. (2015). *See your way to success: Imagery perspective influences performance under stereotype threat* (Publication No. 3710361) [Doctoral dissertation, The Ohio State University]. ProQuest Dissertations and Theses Global.
- *Reinhart, A. L. (2014). *Examining multiple identities under stereotype threat: A regulatory fit perspective* [Doctoral dissertation, University of Texas at Austin]. ProQuest Dissertations and Theses Global.
- Rosette, A. S., & Koval, C. Z. (2018). Framing advantageous inequity with a focus on others: A catalyst for equity restoration. *Journal of Experimental Social Psychology*, 76, 283–289. <https://doi.org/10.1016/j.jesp.2018.03.002>
- Roth, P. L., Bevier, C. A., Bobko, P., Switzer, F. S., III, & Tyler, P. (2001). Ethnic group differences in cognitive ability in employment and educational settings: A meta-analysis. *Personnel Psychology*, 54(2), 297–330. <https://doi.org/10.1111/j.1744-6570.2001.tb00094.x>
- Rothstein, H. R. (2008). Publication bias as a threat to the validity of meta-analytic results. *Journal of Experimental Criminology*, 4(1), 61–81. <https://doi.org/10.1007/s11292-007-9046-9>
- Ryan, A. M., & Sackett, P. R. (2013). Stereotype threat in workplace assessments. In K. F. Geisinger (Ed.), *APA handbook of testing and assessment in psychology* (pp. 661–673). American Psychological Association.
- *Rydell, R. J., McConnell, A. R., & Beilock, S. L. (2009). Multiple social identities and stereotype threat: Imbalance, accessibility, and working memory. *Journal of Personality and Social Psychology*, 96(5), 949–966. <https://doi.org/10.1037/a0014846>
- Sackett, P. R., Borneman, M., & Connelly, B. S. (2008). High stakes testing in education and employment: Evaluating common criticisms regarding validity and fairness. *American Psychologist*, 63(4), 215–227. <https://doi.org/10.1037/0003-066X.63.4.215>
- Sackett, P. R., Hardison, C. M., & Cullen, M. J. (2004). On interpreting stereotype threat as accounting for African American-White differences on cognitive tests. *American Psychologist*, 59(1), 7–13. <https://doi.org/10.1037/0003-066X.59.1.7>
- Sackett, P. R., & Ryan, A. M. (2011). Concerns about generalizing stereotype threat research findings to operational high-stakes testing settings. In T. Schmader & M. Inzlicht (Eds.), *Stereotype threat* (pp. 246–259). Oxford University Press.
- Sackett, P. R., Shewach, O. R., & Keiser, H. N. (2017). Assessment centers versus cognitive ability tests: Challenging the conventional wisdom on criterion-related validity. *Journal of Applied Psychology*, 102(10), 1435–1447. <https://doi.org/10.1037/apl0000236>

- *Salinas, M. F. (1998). *Stereotype threat: The role of effort withdrawal and apprehension on the intellectual underperformance of Mexican-Americans* (Publication No. 9838106) [Doctoral dissertation, University of Texas at Austin]. ProQuest Dissertations and Theses Global.
- *Sanchis-Segura, C., Aguirre, N., Cruz-Gómez, Á. J., Solozano, N., & Forn, C. (2018). Do gender-related stereotypes affect spatial performance? Exploring when, how and to whom using a chronometric two-choice mental rotation task. *Frontiers in Psychology*, 9, Article 1261. <https://doi.org/10.3389/fpsyg.2018.01261>
- *Schmader, T. (2010). Stereotype threat deconstructed. *Current Directions in Psychological Science*, 19(1), 14–18. <https://doi.org/10.1177/0963721409359292>
- Schmader, T., Major, B., & Gramzow, R. H. (2001). Coping with ethnic stereotypes in the academic domain: Perceived injustice and psychological disengagement. *Journal of Social Issues*, 57(1), 93–111. <https://doi.org/10.1111/0022-4537.00203>
- *Shaffer, E. S., Marx, D. M., & Prislín, R. (2013). Mind the gap: Framing of women's success and representation in STEM affects women's math performance under threat. *Sex Roles*, 68(7), 454–463. <https://doi.org/10.1007/s11199-012-0252-1>
- Shewach, O. R., Sackett, P. R., & Quint, S. (2019). Stereotype threat effects in settings with features likely versus unlikely in operational test settings: A meta-analysis. *Journal of Applied Psychology*, 104(12), 1514–1534. <https://doi.org/10.1037/apl0000420>
- *Shih, M., Wout, D. A., & Hambarchyan, M. (2015). Predicting performance outcomes from the manner of stereotype activation and stereotype content. *Asian American Journal of Psychology*, 6(2), 117–124. <https://doi.org/10.1037/a0037707>
- *Smith, J. L., & Johnson, C. S. (2006). A stereotype boost or choking under pressure? Positive gender stereotypes and men who are low in domain identification. *Basic and Applied Social Psychology*, 28(1), 51–63. https://doi.org/10.1207/s15324834basp2801_5
- *Smith, L. G., & Postmes, T. (2011). Shaping stereotypical behaviour through the discussion of social stereotypes. *British Journal of Social Psychology*, 50(1), 74–98. <https://doi.org/10.1348/014466610X500340>
- *Spencer, S. J. (1993). *The effect of stereotype vulnerability on women's math performance* (Publication No. 9332168) [Doctoral dissertation, University of Michigan]. ProQuest Dissertations and Theses Global.
- Spencer, S. J., Logel, C., & Davies, P. G. (2016). Stereotype threat. *Annual Review of Psychology*, 67(1), 415–437. <https://doi.org/10.1146/annurev-psych-073115-103235>
- *Spencer, S. J., Steele, C. M., & Quinn, D. M. (1999). Stereotype threat and women's math performance. *Journal of Experimental Social Psychology*, 35(1), 4–28. <https://doi.org/10.1006/jesp.1998.1373>
- Steele, C. M., & Aronson, J. (1995). Stereotype threat and the intellectual test performance of African Americans. *Journal of Personality and Social Psychology*, 69(5), 797–811. <https://doi.org/10.1037/0022-3514.69.5.797>
- Stone, J., Harrison, C. K., & Mottley, J. (2012). “Don’t call me a student-athlete”: The effect of identity priming on stereotype threat for academically engaged African American college athletes. *Basic and Applied Social Psychology*, 34(2), 99–106. <https://doi.org/10.1080/01973533.2012.655624>
- *Stricker, L. J., & Ward, W. C. (2004). Stereotype threat, inquiring about test takers’ ethnicity and gender, and standardized test performance. *Journal of Applied Social Psychology*, 34(4), 665–693. <https://doi.org/10.1111/j.1559-1816.2004.tb02564.x>
- *Sunny, C. E., Taasoobshirazi, G., Clark, L., & Marchand, G. (2017). Stereotype threat and gender differences in chemistry. *Instructional Science*, 45(2), 157–175. <https://doi.org/10.1007/s11251-016-9395-8>
- *Taasoobshirazi, G., Puckett, C., & Marchand, G. (2019). Stereotype threat and gender differences in biology. *International Journal of Science and Mathematics Education*, 17(7), 1267–1282. <https://doi.org/10.1007/s10763-018-9926-7>
- *Tagler, M. J. (2003). *Stereotype threat: Prevalence and individual differences* (Publication No. 3113977) [Doctoral dissertation, Kansas State University]. ProQuest Dissertations and Theses Global.
- *Tagler, M. J. (2012). Choking under the pressure of a positive stereotype: Gender identification and self-consciousness moderate men's math test performance. *The Journal of Social Psychology*, 152(4), 401–416. <https://doi.org/10.1080/00224545.2011.615353>
- *Tomasetto, C., & Appoloni, S. (2013). A lesson not to be learned? Understanding stereotype threat does not protect women from stereotype threat. *Social Psychology of Education*, 16(2), 199–213. <https://doi.org/10.1007/s11218-012-9210-6>
- Tomeh, D. H., & Sackett, P. R. (2022). On the continued misinterpretation of stereotype threat as accounting for Black–White differences on cognitive tests. *Personnel Assessment and Decisions*, 8(1), Article 1. <https://doi.org/10.25035/pad.2022.01.001>
- *Tsui, M., Xu, X. Y., Venator, E., & Wang, Y. (2016). Stereotype threat and gender: Math performance in Chinese college students. *Chinese Sociological Review*, 48(4), 297–316. <https://doi.org/10.1080/21620555.2016.1166339>
- Turner, J. C., & Reynolds, K. J. (1987). *Rediscovering the social group: A self-categorization theory*. Basil Blackwell.
- Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36(3), 1–48. <https://doi.org/10.18637/jss.v036.i03>
- Walton, G. M., & Cohen, G. L. (2003). Stereotype lift. *Journal of Experimental Social Psychology*, 39(5), 456–467. [https://doi.org/10.1016/S0022-1031\(03\)00019-2](https://doi.org/10.1016/S0022-1031(03)00019-2)
- *Wicherts, J. M., Dolan, C. V., & Hessen, D. J. (2005). Stereotype threat and group differences in test performance: A question of measurement invariance. *Journal of Personality and Social Psychology*, 89(5), 696–716. <https://doi.org/10.1037/0022-3514.89.5.696>

(Appendix follows)

Appendix

Preferred Reporting Items for Systematic Reviews and Meta-Analyses Diagram



Note. See the online article for the color version of this figure.

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